RAFFLES INSTITUTION
2025 YEAR 6 TERM 3 TIMED PRACTICE

CANDIDATE NAME

CLASS 25

COMPUTING 9569

Theory **2 hours**

Additional materials: Answer Papers

READ THESE INSTRUCTIONS FIRST

Answer papers will be provided with the question paper.
Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Student ID

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Answer **all** the questions.
Approved calculators are allowed.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 70.

FOR EXAMINER'S USE					
Q1	Q2	Q3	Q4	Q5	TOTAL
					70

This document consists of **5** printed pages and **1** blank page.

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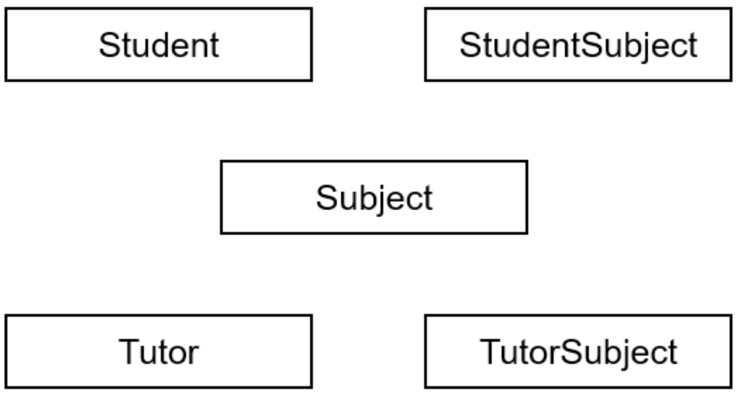
1 A junior college intends to create a centralised consultation booking system and has engaged a small team of H2 Computing students to develop it. The team decides to use a relational database to store relevant information.

First, the database needs to store information about students, including their names, class groups, email addresses and subjects they are taking, as well as information about subject tutors, including their names, departments, email addresses, and teaching information (subjects and the corresponding class groups).

Each subject tutor may be teaching more than one subject, and each subject is to be identified by its unique subject code (e.g. H2 Computing has subject code 9569).

(a) Copy and complete the incomplete entity-relationship (E-R) diagram by adding four relationship lines. [2]



(b) Explain the need to include the `StudentSubject` and `TutorSubject` tables in the relational database. [2]

Next, the database should also store consultation booking information. To book a consultation on the system, the student would need to indicate the tutor, the subject and the consultation timing (e.g. 2024-10-15 15:30).

(c) Two of the tables in the database have the following table descriptions:

`Student(StudentID,StudentName,ClassGroup,StudentEmail)`

`Tutor(TutorID, TutorName, TutorDept, TutorEmail)`

Write table descriptions for the other four tables, ensuring each table is in third normal form and indicating the primary keys and the foreign keys. For these tables, rely on composite keys where appropriate to uniquely identify each record. [7]

(d) State one possible database constraint that can be applied to the booking system to maintain data integrity. [1]

(e) Write an SQL query to find the total number of bookings made per subject, and display the subject name along with the total number of bookings. Order the results by the number of bookings in descending order. [5]

(f) Justify why it is better for the team to develop a web app instead of a native app for the college's consultation booking system. [2]

(g) Is a NoSQL database like MongoDB suited for the same application? Justify your answer. [2]

2 A media access control address (MAC address) is a unique identifier assigned to a network interface card for use as its network address within a network. A typical MAC address looks like this:

`2C:54:91:88:C9:E3`

A MAC address is represented by 12 hexadecimal numbers. For ease of reading, colons are used as a separator between every 2 numbers.

(a) Convert the hexadecimal number `2C:54:91:88:C9:E3` to a binary number, arranging the bits in groups of 4. Show your working clearly. [2]

(b) State the number of bits used to represent the MAC address. [1]

(c) Describe how MAC addresses are used by networking devices during communication within a local area network (LAN). Give one specific example of a situation where MAC addressing is essential. [2]

3 A school is developing a self-service printing kiosk system that allows students to submit print jobs to printers using their student ID and personal device. Students enter their student ID and number of pages through the kiosk interface.

Each printer maintains a queue, printing jobs in the order they are received (first-in, first-out). The system uses Object-Oriented Programming (OOP) to manage printers and print jobs.

For every printer the following data is recorded:

- a unique six-digit ID for the printer
- the model of the printer
- the location of the printer
- the list of print jobs
- the status of the printer (e.g.: "idle", "printing", or "offline").

For every print job the following data is recorded:

- a unique three-digit ID for the print job
- the student ID who submitted the job
- the number of pages to be printed
- the timestamp when the job was submitted (i.e.: 25092024)
- the status of the print job (e.g.: "pending", "in progress", or "completed").

(a) Draw a class diagram showing two classes for the described scenario, showing:

- any derived classes and relationship between the classes
- the properties needed in the classes
- suitable methods, in each class, to support the system.

Include one pair of get and set methods for one of the properties for every class. [8]

(b) Why is it important for the kiosk system to validate user input (e.g, student ID and number of pages) before accepting a print job? [1]

(c) Identify and explain two suitable validation techniques that might be applied to the print job ID. [2]

4 The current self-service printing kiosk system uses a static data structure to implement the print queue. This limits the flexibility of the system because the queue has a fixed size.

(a) Describe how the system could be redesigned using a dynamic data structure. [4]

(b) Explain **two** advantages of selecting a dynamic over static data structure. [4]

(c) Explain a problem that might arise if a dynamic data structure is used in terms of memory allocation. [2]

5 The university intranet can be accessed by all students, both locally and remotely, via the Internet. The network administrator has installed a firewall in the intranet.

(a) Describe two benefits that the intranet might provide for the students. [2]

(b) State the names of two kinds of malware that can infect a particular computer in the university network, and how the infection occurs. [4]

(c) State and explain two ways users can defend their computer against malware. [2]

(d) Describe one other non-malware threat and how it could compromise the university servers. Explain how the servers and the data stored on them could be protected and identify any limitation of the protection. [4]

(e) Most of the students save their work in their laptops. Identify two possible dangers which may lead to the accidental loss of data in the local drives. Describe appropriate strategy to prevent each situation. [4]

(f) Encryption and digital signature are widely used when students and staff send documents to each other. Explain the purpose of using these two in the network security. [2]

(g) The existing network infrastructure is due for replacement and the University plans to provide cloud-based services to the students. Describe three benefits and two risks to the University if the cloud-based approach is adopted. [5]